

(c) Resistor R has a resistance of $56\text{ k}\Omega$. A capacitor of capacitance $12\text{ }\mu\text{F}$ is connected into the circuit of Fig. 6.1 in order to smooth the output voltage.

(i) On Fig. 6.1, draw the capacitor correctly connected into the circuit. [1]

(ii) Calculate the time constant of the smoothing circuit.

time constant = s [2]

(iii) During each discharge cycle, the time for which the capacitor is discharging is 0.14 s .

Determine the minimum value of the smoothed output voltage.

minimum voltage = V [2]

[Total: 15]

